

Julian Louis Ruocco

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Executive Summary

Aerospace Engineering senior graduating Spring 2027 and planning to begin M.S. study in Fall 2027. Seeking a Summer 2027 graduate-level systems engineering internship in space systems, aerospace systems, or mission integration.

Education

The Pennsylvania State University

B.S. Aerospace Engineering

Engineering Leadership Development Minor | Space Systems Engineering Certificate

May 2027

GPA 3.66/4.0

Major GPA 3.82/4.0

Skills

Software & Simulation: MATLAB, Python, HTML, Java, SolidWorks, CAD, STK, COMSOL, KiCad, LabVIEW, AI, Git

Hardware & Fabrication: Control Systems, Microwave Systems, Hardware Integration, Soldering, Welding, Electrical Design

Methodologies: Systems Engineering, Trade Studies, MBSE, Technical Reviews, Technical Writing, Digital Twins, SysML v2

Work Experience

Systems Engineering Intern, Space Dynamics Laboratory, Albuquerque NM

May - July 2026

- Utilized Systems Engineering methodology to iteratively define system requirements and defend a PDR.
- Led ADCS trade studies that converted pointing and attitude-knowledge requirements into reaction-wheel performance targets and star-tracker sensitivity selections for the spacecraft.
- Developed a mission concept for a 6U CubeSat to identify and geolocate ocean plastics in the Pacific Ocean.
- Simulated orbital parameters in STK to determine orbit optimization via imaging pass time and antenna downlink.
- Built a Python script to compress multi-spectral data into indices and used them to identify plastic signatures.

Student Space Programs Laboratory, State College, PA

Laboratory Lead & Chief Engineer

August 2025 - Present

- Led 20 student engineers to win a total of \$125,000 in multiple phases of NASA's LunaRecycle Challenge.
- Designed a novel modular recycling system (RECLAIM) for lunar missions using high power microwaves.
- Defined system requirements with all subsystems leads and integrated them into SysML v2 linked to a Digital Twin which gathered live telemetry from pressure and temperature sensors to validate system requirements.
- Utilized Digital Twin to predict system behavior in off-nominal cases and extraterrestrial environments.
- Integrated and tested RECLAIM under vacuum conditions and temperatures exceeding 500 degrees Celsius.
- Fabricated a proof-of-concept microwave-assisted plastic pyrolysis system with less than \$500 and lab equipment.
- Directed laboratory initiatives, refined the training program, and taught 50+ student engineers systems engineering.

Research Lead, NASA's LunaRecycle Challenge (Phase 1)

December 2024 - May 2025

- Directed the research team for NASA's LunaRecycle Challenge (Phase 1), authoring key proposal sections.
- Created a MATLAB heat transfer model to gauge system temperature, improving energy efficiency by ~15%.

Member, Student Training Program

August 2023 - November 2024

- Applied systems engineering to design and build an atmospheric volatile gas detection payload for a suborbital rocket.
- Designed the electrical wiring of payload in KiCad and integrated the flight hardware onto a circuit board.

Research

Space Systems Researcher, Space Propulsion and Environments Laboratory, State College PA

May - August 2025

- Engineered a technical design and proposal for a lunar recycling system powered by a centralized microwave network.
- Integrated ISRU and catalysts into the design, enabling the recycling of plastics into rocket fuel precursors.